

Information Retrieval – Assignment 1

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System Overview

- Fully Implemented in Python
- Core pipeline:
 - Document parsing & preprocessing (Splitting, stop words removal, etc.)
 - Vectorization
 - Query processing
 - Similarity measure (cosine / TF / TF-IDF / BM25)
 - Evaluation (MAP, P@10)
- Architecture:
 - File_handler.py -preprocessing
 - Eval_handler.py - ranking and scoring
 - Utils.py - pipeline orchestration
 - Run.py - entry point

Baseline (run-0)

- White-space tokenization
- No normalization / stop words removal
- Raw term frequency
- Cosine similarity

Results:

	MAP	P@10
English	0.0413	0.076
Czech	0.050	0.072

Problems:

- Common words noise
- Word variations are treated distinctly

Improvements

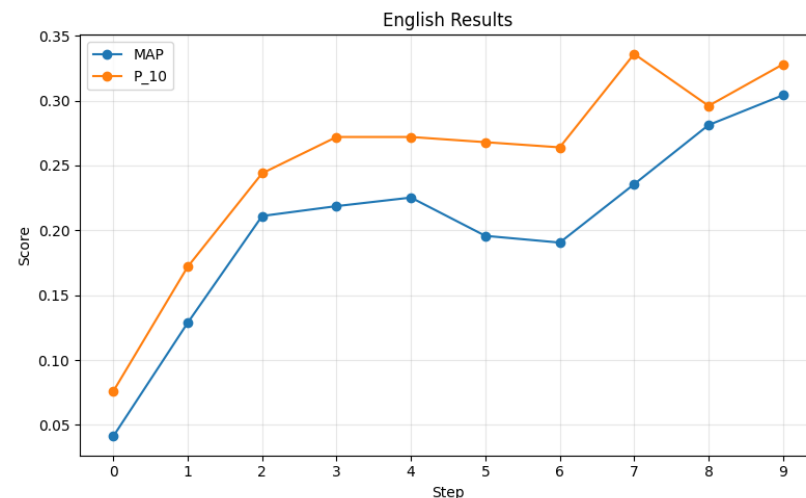
- Improved tokenization
 - Split on lower-case, hyphens, punctuation marks + lowercase
- Stop words removal
- Lemmatization
- Stemming (Only for English)
- TF-IDF weighting
- BM25 ranking (replacing TF-IDF in final system)
- Pseudo relevance feedback (PRF)

Final results run-1 (English: 9 experiments)

- **MAP:** 0.3042
- **P@10:** 0.328

Key insights:

- Tokenization + stop words → biggest gains early
- TF-IDF → major improvement
- BM25 → best MAP
- PRF → final boost



Final results run-1 (Czech: 8 experiments)

- **MAP:** 0.1688
- **P@10:** 0.180

Key insights:

- Lemmatization is crucial
- BM25 underperformed (lacks tuning)
- PRF → helped recovering BM25
- TF-IDF → performed the best



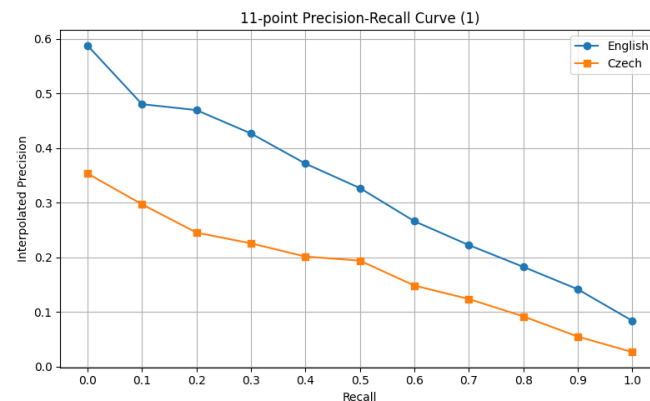
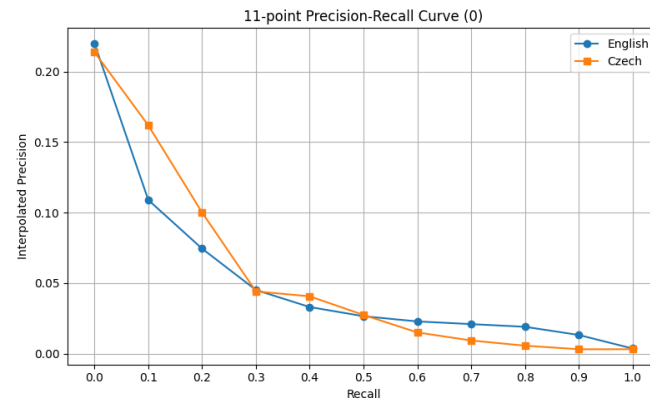
Final comparison & conclusions

- **MAP Improvements**

- English: **+363%** from baseline
- Czech: **+238%** from baseline

- **Takeaways**

- Preprocessing has a huge impact
- TF-IDF and BM25 can have a big impact
 - Requires language-dependent tuning
- Performance is also dependent on language
 - English → simpler morphology
 - Czech → requires strong lemmatization
- IR systems are **not** language independent.



Github repo for full assignment

<https://github.com/OriMoscovitz/Information-Retrieval-Assignment-1.git>